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Thyroid Emergency

Content

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- Myxoedema
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  - Management
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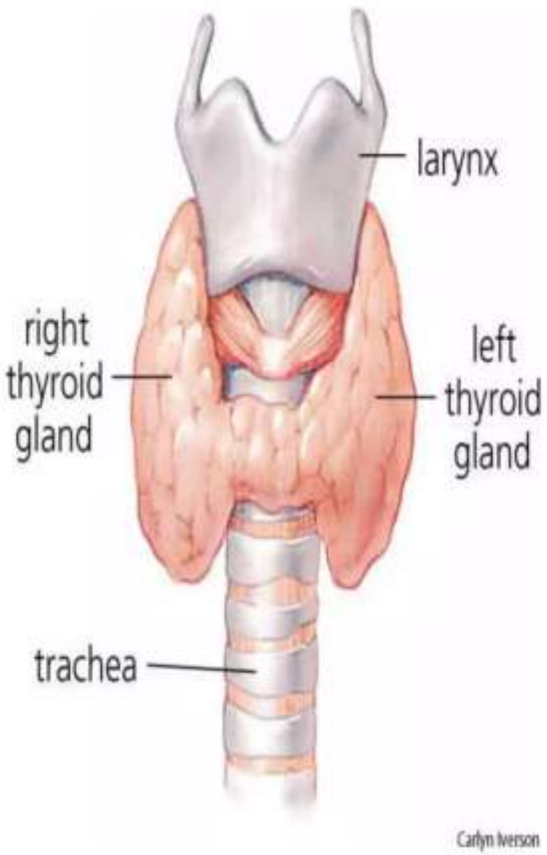
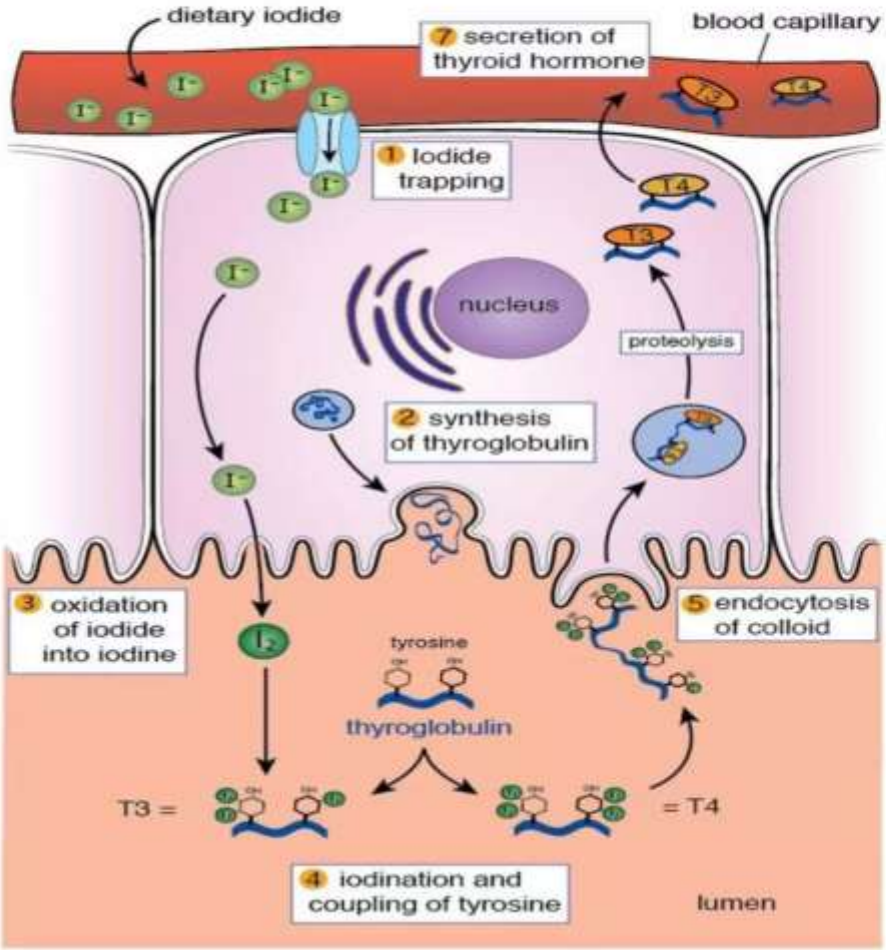
Thyroid Anatomy



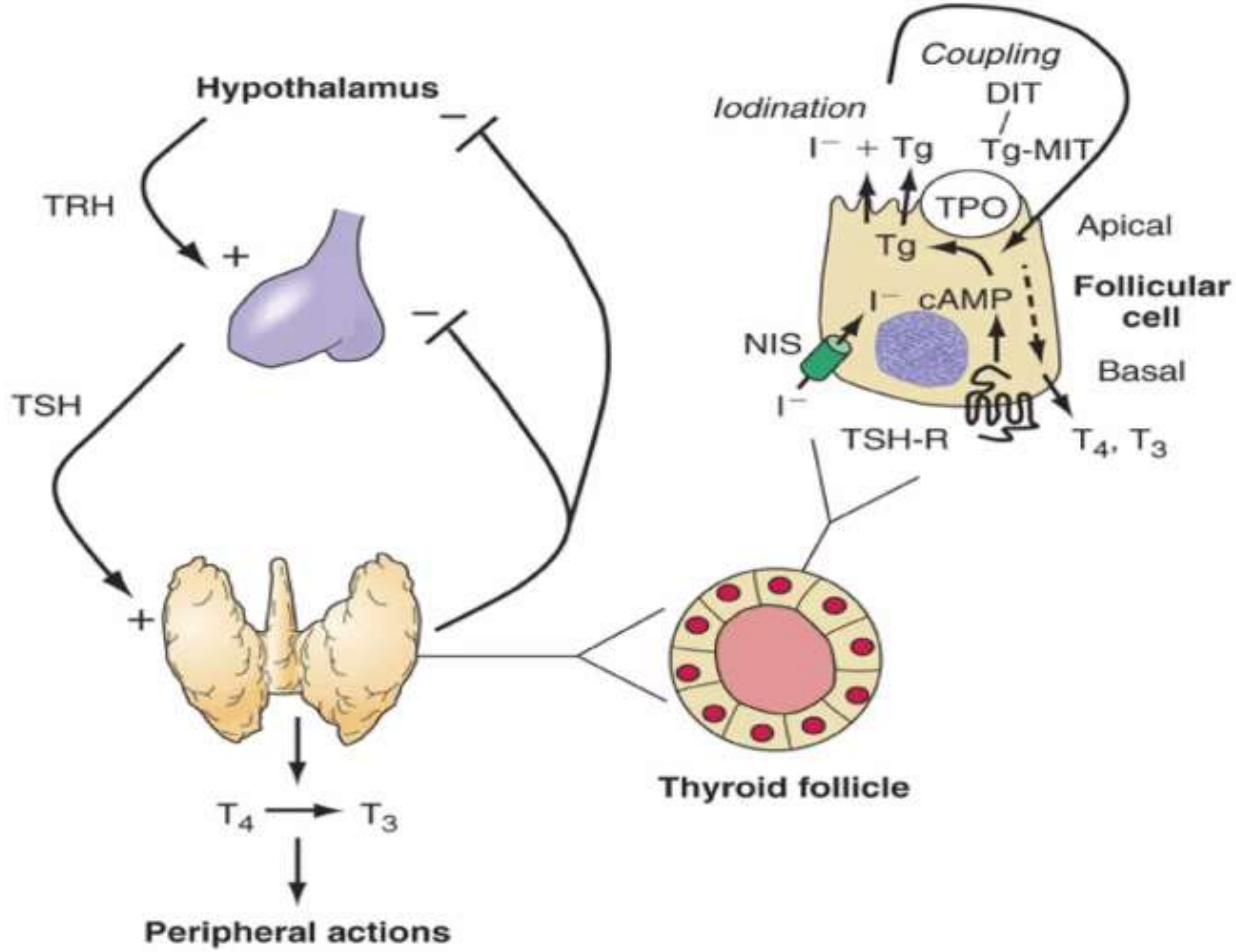
- one of the largest endocrine glands
- consists of two connected lobes
- found in the neck, below the thyroid cartilage (which forms the laryngeal prominence, or "Adam's apple").
- producing thyroid hormones
  - triiodothyronine (T3) and thyroxine tetraiodothyronine (T4)).
  - regulate the growth and rate of function of many other systems in the body.
  - produces calcitonin, which plays a role in calcium homeostasis.

Synthesis and release

- Iodide uptake:Na/I symphorter
- Organification: thyroid peroxidase (TPO) and hydrogen peroxide (HPO)
  - R iodide+thyrosyl residual = di/monothyrosine
- Coupling:
  - DiT+DiT = T4
  - DiT+ MiT= T3
- Storage
- Release



Thyroid hormone regulation



Functions :
Growth, development and metabolism

System	Effect of thyroid hormones
CVS	✓ Increase in heart rate, contractility and cardiac output ✓ Vasodilation
CNS	✓ Mental status alteration ✓ Excitatory effect
Respiratory	✓ Increase in respiration
Gastrointestinal	✓ Increase in GI motility
Endocrine	✓ Increase in secretion and need for other hormones

Thyroid Storm

- Sudden severe life threatening exacerbation of hyperthyroidism associated with multiple organ decompesation
- Mortality rate is 20 – 50 % →FATAL

Etiology

- Most cases secondary to Graves' disease
- Some due to toxic multi-nodular goiter
- Rare causes :
  - Malignancies (most do not efficiently produce thyroid hormones)
- Very rare in children

Precipitating factors

- Infection, especially pneumonia
- Cerebrovascular accident
- Diabetic ketoacidosis
- Major trauma
- Recent surgery
- Iodine 131 Rx or iodine contrast agents
- Rapid withdrawal of anti-thyroid medications

Clinical Presentation

- In any known case of hyperthyroidism with a fever
  - Fever → indicator of any underlying sepsis or consequence of thyroid storm
  - Tachycardia → usually persist during rest/sleep
  - Thyrotoxic symptoms and sign → e.g: weight loss and tremors
  - Multiorgan dysfuctions → CNS, GIT, CVS, Respiratory

Physical Examination

- Hyperpyrexia → underlying sepsis
- Systolic hyper/hypotension, heart failure, atrial fibrillation/flutter
- Tachycardia out of proportion of fever
- Altered mental status → delirium, agitation, stupor, coma
- Volume depletion from fever, increased metabolism, diarrhea
- Stigmata of hyperthyroidism : goitre, tremors, lig lag/retraction, myopathy

Mind the differentials!

- Hypoglycemia
- Hypoxia
- Sepsis
- Encephalitis/meningitis
- Alcohol withdrawal/ drug intoxication
- Heat stroke

Management

- Must be managed in ***critical care*** are due to life threatening nature of the disease
  - Supply HFM oxygen / Venti mask
  - ECG
  - Vital signs every 10-15 mins
  - SPO_2 monitoring
  - 2 large bore peripheral lines

Management

- Administered IV fluids – to correct the volume depletion
- IX: FBC, RP, LFT, Electrolyte, Blood gases, TFT
- Imaging: CXR for evidence of heart failure or infections
- ECG: presence of ischemia/dysarhythmia
- UFEME, DXT
- Correct the percipitating factors
  - Paracetemol , tepid sponging / other cooling technique

DRUGS THERAPHY

[redacted]

Table 7. Three-Step Treatment Of Thyroid Storm

	Goal	Treatment	Effect
Step 1	Block peripheral effect of thyroid hormone	Provide continuous intravenous infusion of β-blocking agent.	Slows heart rate, increases diastolic filling, and decreases tremor.
Step 2	Stop the production of thyroid hormone	Provide antithyroid medication (propylthiouracyl or methimazole) and dexamethasone.	Antithyroids decrease synthesis of thyroid hormone in the thyroid. Propylthiouracil slows conversion of T4 to T3 in periphery. Dexamethasone decreases conversion of T4 to T3 in periphery.
Step 3	Inhibit hormone release	Give iodide 1-2 h after antithyroid medication	Decreases release of thyroid hormone from thyroid.

Drugs Therapy

B-blocker	IV esmolol test dose 250 μg/kg followed by infusion 50μg/min or Iv propranolol 1mg every 5 min until severe tachycardia controlled Oral: - Propanolol 60mg every 4 hours - Propanolol 80 mg every 8 hours
PTU	400-600mg PO stat followed by 200-300mg every 4 h
Iodine	1- 2 hours post PTU therapy - Lugol's iodine 6-8 drops PO/ryle tube every 8h - Nbm: IV sodium iodide 1g/500ml saline every 12h
Corticosteroid	IV Hydrocortisone 100mg every 8h IV Dexamethasone 2mg every 6h

Myxoedema

Myxoedema

- Represents end stage of improperly treated, neglected, or undiagnosed primary **hypothyroidism**
- Occurs in 0.1 % or less of cases of hypothyroidism
- Very rare under age 50
- 50 % of cases become evident after hospital admission
- Mortality is 100 % untreated, 50 % even if treated

Etiology

- Diseases of the :
  - Thyroid (primary hypothyroidism) : 95 %
  - Pituitary (secondary hypothyroidism) : 4 %
  - Hypothalamus (tertiary hypothyroidism) : < 1%
- Can be associated with the multiple endocrine failure syndromes

Clinical Manifestation

- **Neurological symptoms**: confusion, lethargy, psychosis (myxoedema madness), seizures
- **Hypothermia**: impaired thermogenesis
- **Hyponatraemia**: renal impairment/SIADH
- **Hypoventilation**: respiratory acidosis
- **Hypoglycemia**: decreased gluconeogenesis
- **CVS**: bradycardia, heart failure, pericardial effusion, hypotension

Clinical Features

- Signs related to hypothyroidism
  - Fatigue, weakness, cold intolerance, constipation, weight gain, and deepening of voice.
  - Cutaneous signs: dry, scaly, yellow skin, non-pitting, waxy edema of the face and extremities (myxedema): and thinning eyebrows
- CNS:
  - Confusion, lethargy psychosis (myoedema madness) or seizures
- Hypothermia → impaired thermogenesis

Physical Examination

- **General appearances:** altered mental status
- **Vital signs:** Bradycardia, hypotension, hypothermia, hypoventilation
- **CVS:** muffled heart sound, elevated JVP
- **Neurological signs:** focal neurological deficits, tongue laceration (in seizures), slow ankle reflexes
- **Skin:** puffy face and carotinemia
- **Others:** thyroidectomy scars, sepsis evidence

Management

- Blood IX: FBC, RP, CK, ABG, TFT, Cortisol
- DXT
- CXR: look for any cardiomegaly, effusion, pulmonary edema, pneumonia
- ECG

- Supportive management:
  - Vital sign monitoring : Temp, BP, HR, RR, SPO_2
  - IV access and fluid resuscitation
  - Supplemental oxygen
  - Warm with heating blanket
  - IV Hydrocortisone 100mg TDS

- Medications:
  - T3 or T4 (given IV/oral)
  - T3 has rapid onset of action and greater biological activity
- Dose:
  - T3: 2.5µg TDS followed by double dose every 2 or 3 days to target dose of 30-40µg per day
  - T4: 25µg as test dose, then increase to 500µg on first day. Subsequent dosing 25-100µg/day

- Treat the precipitating factors
  - Hypoglycemia : correction with dextrose saline
  - Hyponatremia : slowly infusion of normal saline
  - Cardiac failure : diuretics and vasodilators
  - Sepsis: IV antibiotics preferred

Disposition

- To be admitted to high dependency/ICU unit

Take Home Message

- Thyroid crises masquerade many illness
- Clinical diagnosis is difficult and requires high index of suspicious
- To treat the predisposing cause

	Hypothyroidism (Slow)	Hyperthyroidism (Fast)
Thyroid Function	Underactive	Overactive
Hormone Levels	Low T3, T4	High T3, T4
Metabolism	Slow	Fast
Weight Changes	Gain	Loss
Energy Levels	Fatigue, sluggishness	Hyperactivity, restlessness
Heart Rate	Slow pulse	Rapid heartbeat
Temperature Sensitivity	Cold intolerance	Heat intolerance
Mood Effects	Depression, brain fog	Anxiety, irritability